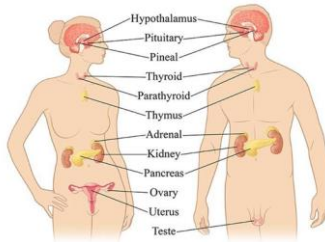


Endocrine system



Cont....

- is a series of **ductless** glands
- secrete hormones into the **circulation**
- allows to act on target cells with receptors for those hormones



- Endocrine cells are **epithelial** in origin
- They are partly contained within **"pure"** endocrine organs and **partly** within organs of other body systems

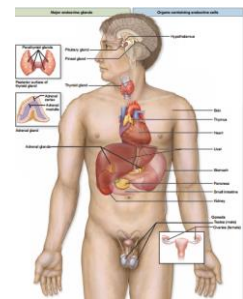
Cont....

- Purely endocrine organs are:
- pituitary gland
- pineal gland
- Thyroid
- parathyroid glands
- adrenal glands



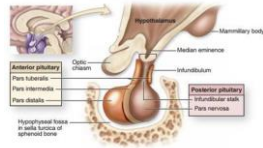
Cont....

- Organs that belong to other body systems but also contain a large proportion of endocrine cells include:
- Pancreas
- thymus
- Gonads
- hypothalamus
- the heart, GIT, kidneys, and skin, also contain scattered or small pockets of cells that secrete hormones



Pituitary gland

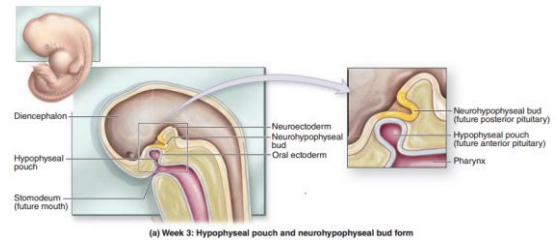
- lies below the brain in a small cavity on the sphenoid bone, the **sella turcica**
- the **infundibulum**, and a **vascular network** connect it to the **hypothalamus**
- has two **functional** components (lobes)
 - Anterior lobe
 - Posterior lobe
- These two portions are of different embryologic origin



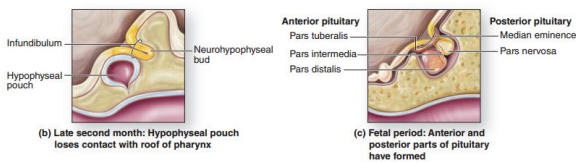
Cont...

I. Anterior lobe

- Adenohypophysis
- derived from an evagination of the ectoderm of the oropharynx toward the brain (Rathke's pouch)

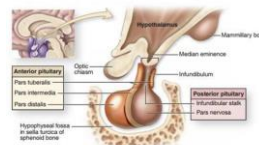


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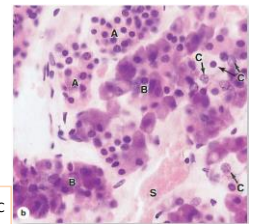


Cont...

- Has three parts
 - Pars Distalis
 - accounts for 75%
 - Has two groups of cells with different staining affinities: chromophils and chromophobes



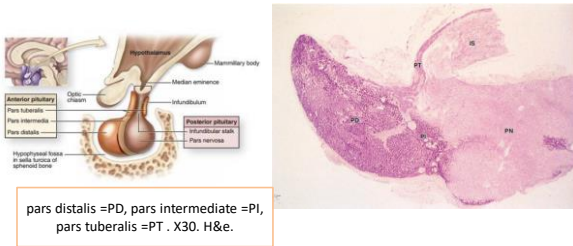
acidophil cells = A basophils = B chromophobes = C
sinusoids = S X400



Cont...

2. Pars tuberalis

- is a smaller funnel-shaped region surrounding the infundibulum of the neurohypophysis
- Most of its cells are **gonadotrophs**



Cont...

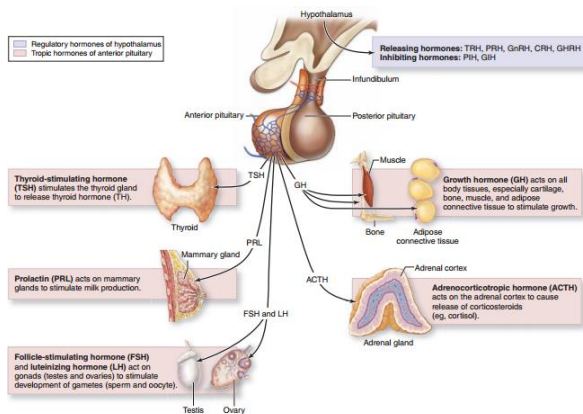
3. Pars intermedia

- is a thin zone of basophilic cells b/n the pars distalis and the pars nervosa



TABLE 20-1 Major cell types of the anterior pituitary and their major functions.			
Cell Type	% of Total Cells	Hormone Produced	Major Function
Somatotrophs	50	Somatotropin (growth hormone, GH), a 22-kDa protein	Stimulates growth in epiphyseal plates of long bones via insulin-like growth factors (IGFs) produced in liver
Lactotrophs (or mammotrophs)	15-20	Prolactin (PRL), a 22.5-kDa protein	Promotes milk secretion
Gonadotrophs	10	Follicle-stimulating hormone (FSH) and luteinizing hormone (LH; interstitial cell-stimulating hormone [ICSH] in men), both 28-kDa glycoprotein dimers, secreted from the same cell type	FSH promotes ovarian follicle development and estrogen secretion in women and spermatogenesis in men; LH promotes ovarian follicle maturation and progesterone secretion in women and interstitial cell androgen secretion in men
Thyrotrophs	5	Thyrotropin (TSH), a 28-kDa glycoprotein dimer	Stimulates thyroid hormone synthesis, storage, and liberation
Corticotrophs	15-20	Adrenocorticotropic hormone (ACTH), a 4-kDa polypeptide Lipotropin (LPH)	Stimulates secretion of adrenal cortex hormones Helps regulate lipid metabolism

TABLE 20-2 Hypothalamic hormones regulating cells of the anterior pituitary.		
Hormone	Chemical Form	Functions
Thyrotropin-releasing hormone (TRH)	3-amino acid peptide	Stimulates release of thyrotropin (TSH)
Gonadotropin-releasing hormone (GnRH)	10-amino acid peptide	Stimulates the release of both follicle-stimulating hormone (FSH) and luteinizing hormone (LH)
Somatostatin	14-amino acid peptide	Inhibits release of both somatotropin (GH) and TSH
Growth hormone-releasing hormone (GHRH)	40- or 44-amino acid polypeptides (2 forms)	Stimulates release of GH
Dopamine	Modified amino acid	Inhibits release of prolactin (PRL)
Corticotropin-releasing hormone (CRH)	41-amino acid polypeptide	Stimulates synthesis of pro-opiomelanocortin (POMC) and release of both β -lipotropin (β -LPH) and corticotropin (ACTH)



Hormones of the Anterior Lobe of the Pituitary Gland

Hormone	Composition	MW (kDa)	Major Functions
Growth hormone (somatotropin, GH)	Straight-chain protein (191 aa)	21,700	Stimulates liver and other organs to synthesize and secrete insulin-like growth factor I (IGF-I), which in turn stimulates division of progenitor cells located in growth plates and in skeletal muscles, resulting in body growth
Prolactin (PRL)	Straight-chain protein (198 aa)	22,500	Promotes mammary gland development; initiates milk formation; stimulates and maintains secretion of casein, lactalbumin, lipids, and carbohydrates into the milk
Adrenocorticotropic hormone (ACTH)	Small polypeptide (39 aa)	4,000	Maintains structure and stimulates secretion of glucocorticoids and gonadocorticoids by the zona fasciculata and zona reticularis of the adrenal cortex
Follicle-stimulating hormone (FSH)	2-chain glycoprotein* (α , 92 aa; β , 111 aa)	28,000	Stimulates follicular development in the ovary and spermatogenesis in the testis

Luteinizing hormone (LH)	2-chain glycoprotein* (α , 92 aa; β , 116 aa)	28,300	Regulates final maturation of ovarian follicle, ovulation, and corpus luteum formation; stimulates steroid secretion by follicle and corpus luteum; in males, essential for maintenance of and androgen secretion by the Leydig (interstitial) cells of the testis
Thyrotropic hormone (TSH)	2-chain glycoprotein* (α , 92 aa; β , 112 aa)	28,000	Stimulates growth of thyroid epithelial cells; stimulates production and release of thyroglobulin and thyroid hormones

II. Posterior lobe

- Neurohypophysis
- is an extension of the central CNS that stores and releases secretory products from the **hypothalamus**
- consists of **pars nervosa** and the **infundibulum**
- Contain:
 - 100,000 unmyelinated axons of large secretory neurons with cell bodies in the **supraoptic** and **paraventricular** nuclei of the hypothalamus
 - glial cells called **pituicytes**

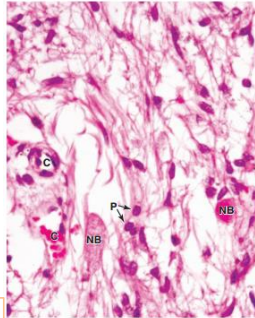


Cont....

Cont....

- Hypothalamic-Hypophyseal Tract (axonal transport)
- store two hormones:
 - antidiuretic hormone (ADH)
 - Oxytocin hormones
- accumulate in axonal dilations called **neurosecretory bodies** or **Herring bodies**

pituicytes = P neurosecretory (Herring) bodies
= NB capillaries = C X400. H&E.



Hormone (Cell Type)	Target Organ and Effects	Effects of Hyposecretion ↓ and Hypersecretion ↑
POSTERIOR LOBE HORMONES (MADE BY HYPOTHALAMIC NEURONS AND STORED IN POSTERIOR LOBE)		
 Oxytocin (from neurons in paraventricular nucleus of hypothalamus)	 Uterus: stimulates uterine contractions; initiates labor; breast: initiates milk ejection	Unknown
Antidiuretic hormone (ADH) or vasopressin (from neurons in supraoptic nucleus of hypothalamus)	 Kidneys: stimulates kidney tubule cells to resorb water	↓ Diabetes insipidus ↑ Syndrome of inappropriate ADH secretion (SIADH)

Cont....

Cont...

Blood Supply Pituitary gland

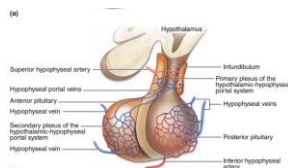
- branches of Internal carotid

1. Superior hypophyseal arteries

- supply the pars tuberalis, median eminence, and infundibulum
- hypothalamohypophyseal portal system provides link b/n the hypothalamus and the pituitary

2. Inferior hypophyseal arteries

- supply the pars nervosa

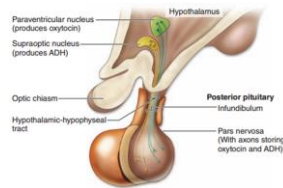


disorders of the pituitary glands?????

- problems caused by tumors fall into certain categories:
 - Hyposecretion
 - Hypersecretion
 - Mass effects

Hypothalamus

- it is the **neuroendocrine** gland
- coordinates most **endocrine** functions and serves as one of the major controlling centers of the **autonomic** nervous system
- In addition to **oxytocin** and **ADH**, hypothalamic neurons secrete polypeptides that promote and inhibit the secretion and release of hormones from the anterior lobe of the pituitary gland
- found towards the **base** of the brain, below the thalamus



Hormone	Chemical Form	Functions
Thyrotropin-releasing hormone (TRH)	3-amino acid peptide	Stimulates release of thyrotropin (TSH)
Gonadotropin-releasing hormone (GnRH)	10-amino acid peptide	Stimulates the release of both follicle-stimulating hormone (FSH) and luteinizing hormone (LH)
Somatostatin	14-amino acid peptide	Inhibits release of both somatotropin (GH) and TSH
Growth hormone-releasing hormone (GHRH)	40- or 44-amino acid polypeptides (2 forms)	Stimulates release of GH
Dopamine	Modified amino acid	Inhibits release of prolactin (PRL)
Corticotropin-releasing hormone (CRH)	41-amino acid polypeptide	Stimulates synthesis of pro-opiomelanocortin (POMC) and release of both β -lipotropin (β -LPH) and corticotropin (ACTH)

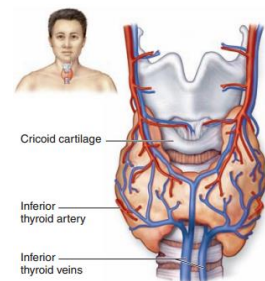
Cont...

Hypothalamic Disease????????

- A disease or disorder of the hypothalamus is known as a hypothalamic disease

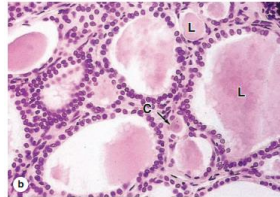
Thyroid gland

- Is the body's **largest** endocrine gland
- develop from the **endodermal** lining of the floor of the primitive pharynx
- lies deep to the **sternothyroid** and **sternohyoid** muscles
- Composed of two lobes connected by an **isthmus**
- Surrounded by a thin fibrous capsule
- supplied by the **superior**, **inferior** thyroid arteries and **thyroid ima** artery (10% of people)



Cont....

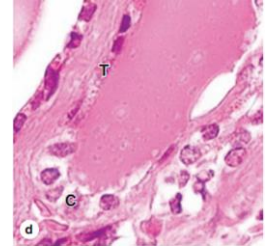
- It synthesizes the thyroid hormones **thyroxine** (*tetra-iodothyronine* or T4) and *triiodothyronine* (T3)
- T4 and T3 help control the basal metabolic rate in cells throughout the body
- as well as the polypeptide hormone **calcitonin**
- thyroid **follicle** is the structural and functional unit of the thyroid gland



lumen = L of each follicle
parafollicular or C cells = C X200.
H&E.

Cont....

- Follicular epithelium contains **follicular** and **parafollicular** cells
- **Follicular** cells (principal cells) are responsible for production of the **thyroid hormones** T4 and T3
- **Parafollicular** cells secrete calcitonin, a hormone that regulates **calcium** metabolism



Thyrocytes = T parafollicular C cells = C X400.
H&E;

Abnormal thyroid function???

Hormones of the Thyroid Gland			
Hormone	Composition	Source	Major Functions
Thyroxine (tetraiodothyronine, T4) and triiodothyronine (T3)*	Iodinated tyrosine derivatives	Follicular cells (principal cells)	Regulates tissue basal metabolism (increases rate of carbohydrate use, protein synthesis and degradation, and fat synthesis and degradation); regulates heat production; influences body and tissue growth and development of the nervous system in the fetus and young child*; increases absorption of carbohydrates from the intestine
Calcitonin (thyrocalcitonin)	Polypeptide containing 32 amino acids	Parafollicular cells (C cells)	Decreases blood calcium levels by inhibiting bone resorption and stimulating absorption of calcium by the bones



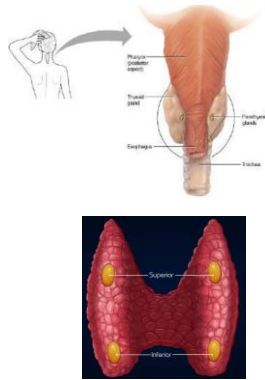
(c) Goiter (enlargement of thyroid gland)



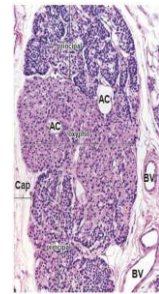
young woman with signs of hyperthyroidism

Parathyroid gland

- Small & pea shaped glands
- Two in each lobe; superior & inferior embedded in the posterior wall of the fibrous capsule of **thyroid** gland
- develop from the endodermal cells derived from the **third** and **fourth** **branchial** pouches



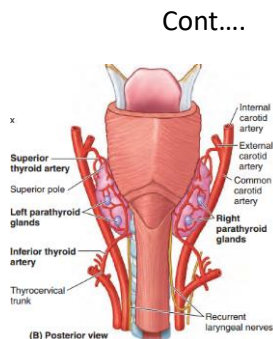
- **Principal** cells and **oxyphil** are the epithelial cells of the parathyroid gland
- Principal (chief) cells are responsible for regulating the synthesis, storage, and secretion of large amounts of **parathyroid hormone**
- Oxyphil cells constitute a minor portion of the parenchymal cells and are not known to have a **secretory** role
- Parathyroid hormone regulates **calcium** and **phosphate** levels in the blood



human parathyroid gland. H&E. X 175.
connective tissue capsule = **Cap**
blood vessels = **BV**
Adipose cells = **AC**

Cont....

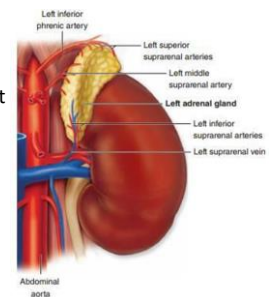
- **inferior thyroid arteries** (primary)
- branches from the superior thyroid arteries; thyroid ima artery; or laryngeal, tracheal, and esophageal arteries
- **Parathyroid veins** drain into the **thyroid plexus of veins**
- **Lymphatic vessels** into deep cervical lymph nodes and paratracheal lymph nodes
- nerve supply is derived from thyroid branches of the cervical (sympathetic) ganglia



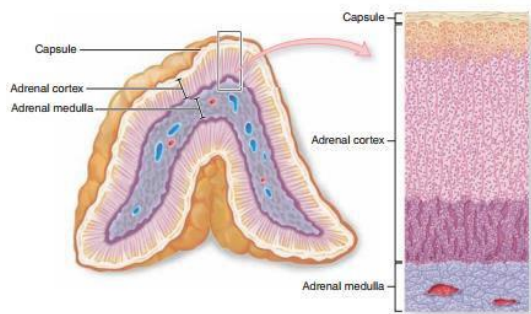
Cont....

Adrenal or suprarenal gland

- located b/n the superomedial aspects of the **kidneys** and the **diaphragm**
- Each suprarenal gland has :
 - the suprarenal cortex and suprarenal medulla with different **embryological** origins and functions
 - cortical cells originate from **mesodermal mesenchyme**, whereas the medulla originates from **neural crest** cells



Cont....



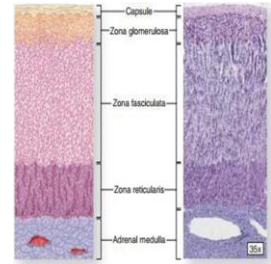
I. adrenal cortex

- cells have acidophilic cytoplasm rich in lipid droplets with central nuclei

- has three concentric zones:

1. **Zona glomerulosa**

- immediately inside the capsule
- Comprise about 15% of the cortex
- synthesize and secrete **mineralocorticoids**
- control the amount of minerals in the blood



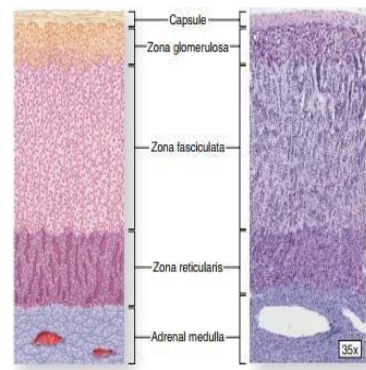
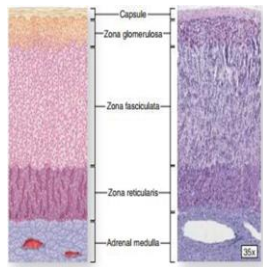
Cont....

2. **Zona fasciculata**

- occupies 65% to 80% of the cortex
- synthesize and secrete **glucocorticoids**
- maintain normal glucose concentration

3. **zona reticularis**

- Innermost comprising about 10% of the cortex
- synthesize and secrete small amount of sex hormones

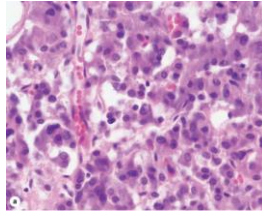


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Cont....

B. Adrenal medulla

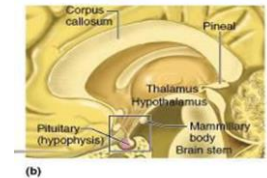
- Contains chromaffin cells
- arise from neural crest cells
- secrete the catecholamine (epinephrine and norepinephrine)
- Catecholamines released in response to intense emotional reaction, important for fight or flight activities



. X200. H&e.

Pineal gland (pineal body)

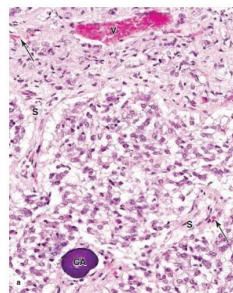
- epiphysis cerebri
- is a small, pine cone shaped structure on the roof of the **diencephalons**
- it also influences reproductive system and gonadal activities
- it's activity influenced by daily cycle of light and dark



(b)

Cont....

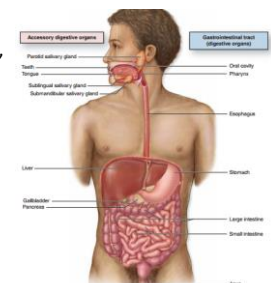
- composed of cells called **pinealocytes** which secrete **melatonin**
- also has interstitial **glial** cells that are modified astrocytes, which represent about 5% of the cells
- concretions of calcium and magnesium salts called **corpora arenacea** are characteristic feature of the gland.



septa = S, venules = V, capillaries = arrows,
Corpus arenaceum = CA X200. H&e

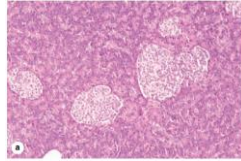
The pancreas

- in the upper abdomen, just under the **stomach**
- The major part of the pancreas, called the **exocrine** pancreas functions as an exocrine gland, secreting digestive enzymes in to GIT
- The endocrine cells of the pancreas are contained in spherical bodies called **pancreatic islets** or **islets of Langerhans**



Cont....

- It has more than 1 **million** islets, mostly in the gland's narrow tail region,
- but they only constitute 1% to 2% of the organ's total volume
- Are epithelial outgrowths from **endoderm** of the developing gut



. X40. H&e.

Cell Type	Quantity (%)	Hormone Produced	Hormone Structure and Size	Hormone Function
α	~20	Glucagon	Polypeptide; 3500 Da	Acts on several tissues to make energy stored in glycogen and fat available through glycogenolysis and lipolysis; increases blood glucose content
β	~70	Insulin	Dimer of α and β chains with S-S bridges; 5700-6000 Da	Acts on several tissues to cause entry of glucose into cells and promotes decrease of blood glucose content
δ or D	5-10	Somatostatin	Polypeptide; 1650 Da	Inhibits release of other islet cell hormones through local paracrine action; inhibits release of GH and TSH in anterior pituitary and HCl secretion by gastric parietal cells
PP	Rare	Pancreatic polypeptide	Polypeptide; 4200 Da	Stimulates activity of gastric chief cells; inhibits bile secretion, pancreatic enzyme and bicarbonate secretion, and intestinal motility

The gonads

- Gonads are organs that produce gametes
 - sperm in males
 - oocytes in female
- **female sex glands**
 - ovaries**
- are located in the pelvic cavity
- composed of a **cortex** and a **medulla**



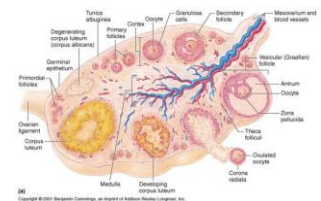
Cont....

1. ovarian follicles

- secrete **estrogen**

2. corpus luteum

- chiefly secretes **progesterone** and some **estrogen**
- estrogens maintain the reproductive organs and secondary sex characteristics of females

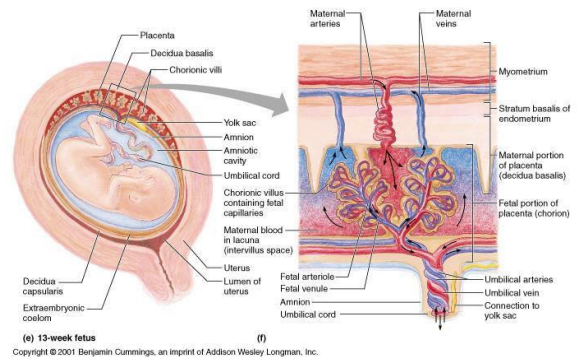


Thymus

- located in the lower **neck** and anterior thorax (mediastinum)
- In infants it may extend up to lower edge of thyroid gland as age increases it will atrophy
- Secretes a hormone called **thymosin** and **thymopoietin**
- which causes T lymphocytes to gain immunocompetence (important for the development and functioning of the body's immune system)

Placenta

- is a **temporary** endocrine organ formed during pregnancy, which produces hormones important in the maintenance of a healthy pregnancy and in preparation for labour and breastfeeding
- estrogen , progesterone and human choriionic gonadotrophin (HCG)



Diffuse neuroendocrine system (DNES)

- These secretory cells are considered neuroendocrine because they produce many of the same polypeptides and
 - **neurotransmitter-like** molecules, such as serotonin (5-hydroxytryptamine), also released by neurosecretory cells in the CNS
- Some of the hormones signal nearby target cells without first entering the blood stream
- The **GIT** and its derivatives: enteroendocrine cells are hormone-secreting scattered epithelial cells lining the alimentary canal and related endocrine cells, **respiratory** tubes, **pancreas**

Cont...

- Other endocrine structures

heart

- the atria secrete atrial natriuretic peptide (ANP), stimulates kidney to increase its secretion of salt and its production of salty urine

kidneys

- juxtaglomerular apparatus secretes renin, signals adrenal cortex to secrete aldosterone
- other kidney cells secrete erythropoietin signals the bone marrow to increase production of red blood cells

Skin

- is the largest single organ of the body
- accounting for 15% to 20% of total body weight
- performs secrete hormones, cytokines, and growth factors
- convert precursor molecules into hormonally active molecules (vitamin D)